The Log-Normal Distribution

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Introduction

If $\log X$ is normally distributed, then X is log-normal. The distribution is strictly positive and right-skewed, and it models variables that result from multiplicative rather than additive processes – concentrations, income, survival times, and many biomedical lab values.

Prerequisites

Normal distribution, basic properties of logarithms.

Theory

A random variable $X > 0$ is log-normal with parameters μ and σ (on the log scale) if $\log X \sim \mathcal{N}(\mu, \sigma^2)$.

PDF:

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log x - \mu)^2}{2\sigma^2}\right\}, \qquad x > 0.$$

Moments:

- $E[X] = \exp(\mu + \sigma^2/2)$.
- $\mathrm{Var}(X) = \left[\exp(\sigma^2) - 1\right] \exp(2\mu + \sigma^2)$.
- Median $= e^{\mu}$; mode $= e^{\mu - \sigma^2}$.

The **geometric mean** of a log-normal sample equals e^{μ}, the median; it is the natural point estimate when the data combine multiplicatively.

Multiplicative CLT: the product of a large number of independent positive random variables tends to log-normal (by applying the CLT to the sum of logs).

Assumptions

Strictly positive continuous data whose logarithm is approximately normal.

R Implementation

```r
mu    <- 1.0
sigma <- 0.5

X <- rlnorm(1e5, meanlog = mu, sdlog = sigma)
c(mean_X          = mean(X),
  theoretical     = exp(mu + sigma^2 / 2),
  median_X        = median(X),
  theoretical_med = exp(mu))

# Log-transform makes it normal
```

```r
hist(log(X), breaks = 50, freq = FALSE, col = "#2A9D8F")
curve(dnorm(x, mu, sigma), add = TRUE, col = "#F4A261", lwd = 2)

# Geometric mean equals median (for lognormal)
gm <- exp(mean(log(X)))
c(geometric_mean = gm, median = median(X), theoretical = exp(mu))

# MLE
est_mu    <- mean(log(X))
est_sigma <- sd(log(X))
c(mu_hat = est_mu, sigma_hat = est_sigma)
```

Output & Results

```
      mean_X   theoretical      median_X  theoretical_med
       3.082         3.080         2.716            2.718

geometric_mean    median   theoretical
         2.714     2.716         2.718

   mu_hat    sigma_hat
    0.9995       0.500
```

Empirical arithmetic mean is larger than the median (right skew), the geometric mean equals the median (a property of log-normals), and the log-scale MLEs recover the true μ and σ.

Interpretation

For log-normal variables (concentrations, titres), report the geometric mean and a geometric-mean-based confidence interval rather than the arithmetic mean and SD. In a manuscript: "geometric mean antibody titre 588 (95 % CI 441-782)".

Practical Tips

- Plot on a log scale when visualising log-normal data; the skew is removed.
- For regression on log-normal outcomes, log-transform the response and fit a linear model on the log scale.
- Back-transformation of the mean of the log-transformed variable gives the geometric mean, not the arithmetic mean.
- For inference on the arithmetic mean, use $\hat{\mu} + \hat{\sigma}^2/2$ on the log scale and back-transform, but this requires a bias correction in small samples.
- Zero or negative values cannot be log-transformed; substitute or switch to a zero-inflated or Tobit model.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots